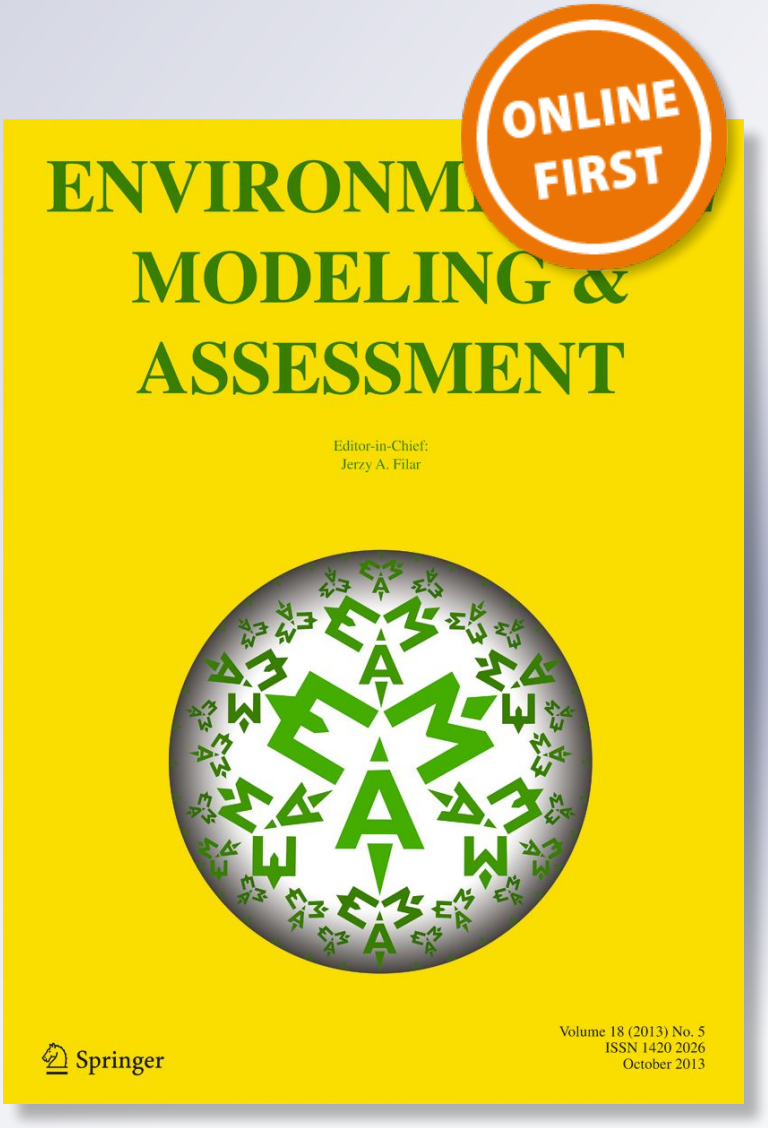
Monetizing the Value of Measurements of Equilibrium Climate Sensitivity Using the Social Cost of Carbon

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ORIGINAL PAPER

Monetizing the Value of Measurements of Equilibrium Climate Sensitivity Using the Social Cost of Carbon

Roger M. Cooke[1] · Alexander Golub[1] · Bruce Wielicki[1] · Martin Mlynczak[1] · David Young[1] · Rosemary Rallo Baize[1]

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Abstract
There has been much work on the value of learning about climate and the value of information regarding the climatic system. The present research moves beyond an abstract hypothesis about future learning and considers concrete Earth Observing Systems that could enhance knowledge of the climatic system and better inform decision makers. This study shows how real options theory in combination with the social cost of carbon may help to calculate the value of information regarding equilibrium climate sensitivity and estimate the relative advantages of two different Earth Observing Systems (EOSs) for learning about ECS. One system aims to improve measurements of decadal global temperature increase and another targets measurements of decadal change of global cloud radiative effect. The paper concludes that a new EOS that substantially reduces the uncertainty in cloud radiative effect would be expected to have more value than improving estimations of the global surface temperature alone.

Keywords Value of information · Climate sensitivity · Real options · Social cost of carbon

1 Introduction

Equilibrium climate sensitivity (ECS) is a commonly used metric for the warming of the Earth's surface induced by doubling the atmospheric concentration of CO_2. This study shows how real option theory in combination with the Interagency Working Group on Social Cost of Carbon ([13, 14] and IWGSSC 2013, denoted here SCC) may be used to monetize the value of two different Earth Observing Systems (EOSs) for learning about ECS. One measurement program targets the decadal rate of change of global surface temperature (GST), the other focuses on decadal percentage change in shortwave cloud radiative effect (CRE).[1] The analysis builds on Cooke et al. [1–3] and Wielicki et al. [31].

Recent reports of the National Academy of Science [23] and the fifth assessment report of the IPCC emphasize the importance of transient climate response (TCR: the global temperature response after 70 years from increasing emissions by 1% per year) for policy decisions. The EOSs described here actually measure aspects of TCR; however, in the current social cost of carbon (SCC) methodology, there is a unique relation between the TCR and ECS. The calculations underlying the results presented here use the current SCC methodology and refer to ECS rather than TCR. This does not constitute an endorsement of the models and assumptions in the current SCC methodology, but rather an acknowledgement that a better methodology is not yet available. All damage calculations refer to the transient regime and make no appeal to an equilibrium warming, beyond the fact that transient and equilibrium behaviors are coupled in Dynamic Integrated Climate–Economy (DICE). An appendix explains shortfalls in the current carbon cycle modeling and shows how the new model proposed by NAS (2017) yields better performance.

There has been much work on the value of learning about climate and the value of information [16, 20, 27, 30]. This literature generally views learning as learning the value of some physical variables with certainty at some future time and computing the benefits. Hope [11] considers a specific reduction of uncertainty at given times. Based on the PAGE09 integrated assessment model (which is also used in the Interagency

[1] Cloud radiative effect or CRE has also often been called cloud radiative forcing (CRF). The quantities are identically defined. We use CRE in this paper as it has become the more common usage

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Working Group on the Social Cost of Carbon IWGSCC), he reports: "Approximately halving the uncertainty range for TCR [Transient Climate Response] has a net present value of about $10.3 trillion (year 2005 US$) if accomplished in time for emissions to be adjusted in 2020, falling to $9.7 trillion if accomplished by 2030." Hope [11] does not consider *how* the learning or uncertainty reduction will be achieved, though he does suggest that the learning will result from high-resolution climate models run on super computers.

The present research moves beyond an abstract hypothesis about future learning and considers concrete Earth Observing Systems. A new proposed enhanced EOS, and the current EOS it is designed to replace, must acquire an uncertainty profile, whereby the variance in its measurement results is decomposed as a sum of variances resulting from natural variability, which cannot be removed by measurement and instrumental uncertainty, which can be reduced by better instruments. When the EOSs are profiled in this way, it becomes evident that (1) we never learn with certainty, we can only reduce uncertainty, and (2) regarding climate trends, *when* we learn depends on the length of the observation period, the measurement accuracy, and the unknown value of the underlying physical quantity. In this context, it is not appropriate to posit a given degree of certainty attained at a given time.

Uncertainty has a physical dimension; a standard deviation has the units of the underlying variable, meters per second, grams per unit area, etc. Therefore, simply knowing the uncertainty of a measurement system tells us little unless we know the scale of the thing we want to measure. Further, the value of a new measurement must be assessed in terms of its improvement over existing measurement systems. If different temporal trends are evidence of unknown physical quantities of interest, we must have a common metric for comparing measures of the different trends. Finally, a measure for the value of learning the unknown quantities is required. Only then can we quantify the value which the alternative measurements could realize.

All of these threads come together in comparing two recently proposed EOS targeting GST rise and CRE trends, as indicative of ECS. This study shows how the social cost of carbon (SCC) methodology enables treating the EOS's for GST and CRE as real options and quantifying their potential contributions as real option values (ROVs).

Learning does not guarantee properly acting on the newly acquired information. Decision makers have merely an option to act choosing the best possible strategy given the new information. Application of ROV allows calculation of incremental benefits of learning but does not predict actual actions of policy makers. Irrational or sub-optimal behavior of policy makers does nullify the value they have chosen to forego. In addition, it is impossible to predict exactly what conditions will trigger global action to reduce climate change. One of the key results of these and previous studies [1] is that whatever observations are needed to reduce uncertainty in future climate change and its impacts, those levels of certainty will be achieved 15 to 30 years sooner using more advanced climate change observations. Cooke et al. (2014) further showed low sensitivity to the specific chosen societal decision triggers. Large changes in confidence for a climate change signal needed to trigger a societal decision (e.g., 80% vs 99% confidence) led to only modest changes in the economic value. This result was obtained despite the fact that these confidence changes changed the timing of when society acted by up to a few decades. The primary economic value result was robust to uncertainty in society's method of triggering action.

Economic value of information studies can also assist in decisions about which observations are more or less effective in supporting societal decisions. An example is that National Academies of Sciences, Engineering, and Medicine 2017 NASA's CLARREO mission was forced to choose between advancing accuracy in thermal infrared versus reflected solar observations due to limited budgets for mission development. The mission chose the reflected solar observations over infrared, primarily on their perceived larger societal value.

The analyses given here can ultimately support a much larger effort to design and implement a global climate observing system [33]. Showing that heightened measurement accuracy of important climate variables yields real social benefits sets the stage for valuing a wide range of critical measurements targeting sea level rise, ocean heat storage, aerosol forcing, and many others [22]. Such valuation is critical in standing up a complete climate observing system.

2 Measure for Measures

Most climate observations are obtained from observing systems designed for other functions such as weather observations (NOAA and EUMETSAT satellites) or research observations (NASA or ESA satellites). This often means that the accuracy of such systems is less than optimal for measuring long-term decadal climate change. The NRC Decadal Survey for Earth Science [21] concluded that accuracy remains a major challenge for climate change observations. That report recommended an in-orbit reference calibration space mission designed to achieve factors of 5 to 10 improvement in accuracy traceable to international standards (SI). The traceability to international standards enables improved accuracy in climate change trends by greatly reducing uncertainty of instrument calibration drifts over long periods of time in orbit, as well as absolute calibration shifts across data gaps. The calibration mission is named CLARREO (Climate Absolute Radiance and Refractivity Observatory). CLARREO is designed to allow intercalibration of orbiting radiometers and spectrometers observing the Earth across the entire reflected solar and Earth-emitted infrared spectrum [21]. A summary of

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the mission design can be found in Wielicki et al. [31]. More than 100 Earth viewing sensors in orbit could have their calibration improved by reference calibration orbiting systems such as CLARREO [31] or a UK proposed calibration mission called TRUTHS (Fox et al. 2011), although the latter involves only reflected solar radiation, not longwave radiation. The critical importance of improved calibration for climate change observations again confirmed in two more recent National Academy reports [22, 25].

High accuracy reference calibration missions similar to CLARREO and TRUTHS have been called for by the Global Space Based Inter-Calibration System (GSICS) [9] as well as the Global Climate Observing System Implementation Plan [8], both of which are part of the World Meteorological Organizations World Climate Research Program. Such missions have also been called for by the report on "Strategy Towards an Architecture for Climate Monitoring from Space" (2013) produced jointly by the World Meteorological Organizations, the Committee on Earth Observing Satellites, and the Coordination Group for Meteorological Satellites.

Recently, a U.S. Academy of Sciences report titled "Continuity of NASA Earth Observations from Space: A Value Framework" took a major step towards quantifying climate science objectives such as uncertainty in ECS and showed the relationship between that quantification and increased accuracy in climate change observations [25].

Despite these strong endorsements for more accurate global satellite observations of climate change, and despite the demonstrated readiness of technology to achieve reference calibration instruments in orbit [17, 31], these advances remain grounded in the laboratory. Part of the challenge is that the climate system is much larger and more complex than weather, with about 10 times more essential climate variables than weather variables (GCOS, 2017). Further, because decadal climate signals are much smaller than daily weather change signals, accuracy requirements are typically an order of magnitude more stringent for climate observations (NRC, 2015, [5, 21]). This provides one of the motivations in the present paper to better understand the economic value of increased accuracy in climate change trends. It also suggests the high importance of understanding the relative value of different climate observations. We use this motivation to explore and compare in this paper the relative economic value of calibration improvements for two key climate change observations relevant to narrowing uncertainty in climate sensitivity.

In 2017, NASA began formal development of a reflected solar CLARREO Pathfinder mission to demonstrate both the improved in orbit accuracy and the ability to transfer that improvement to other key space orbiting instruments used for climate change observations. The mission is planned for launch to the international space station in late 2020 as a class D mission with a budget under $100 million US 2017 dollars. This first demonstration mission will cover only the reflected solar part of the Earth's spectrum from 350 to 2300 nm wavelengths. An infrared companion demonstration mission has not yet been started, although laboratory instruments have demonstrated the ability to reach the necessary levels of calibration accuracy. The CLARREO Pathfinder mission would be relevant to the CRE climate trends discussed in this paper, while a future infrared mission would be relevant to the GST climate trends. Having both reflected solar and infrared instruments in orbit simultaneously is obviously highly preferred, a subject to which we return in the concluding section. While CLARREO is used as a quantitative example of an improved climate observing system, it is only one key part of such a future system.

3 Inferring ECS from GST and CRE

According to the climate model implemented in the integrated assessment model DICE [24, 26] as certified by the SCC, there is a unique relation between the transient decadal rate of global surface temperature rise and ECS for a given emission path. Under the business as usual (BAU) emission path, this relation is depicted in Fig. 1.

The relation between ECS and CRE is mediated by cloud feedbacks. Decadal change in global mean shortwave cloud radiative effect (CRE) is linearly related to the magnitude of low cloud feedback, which is the dominant uncertainty for climate sensitivity [4, 12, 29, 34]. Shortwave or reflected solar CRE dominates low cloud feedback in the climate system because the cloud temperature for these cloud systems is very close to surface temperature, thereby greatly limiting their impact on longwave or thermal infrared cloud feedback. Total cloud feedback is the sum of longwave and shortwave feedback. As a result, we focus in this paper on uncertainty in decadal change observations of global mean shortwave CRE as a measure most directly linked to uncertainty in climate sensitivity. Observations of shortwave CRE are derived from the difference of all-sky and clear-sky top of atmosphere radiative fluxes observed by broadband radiation satellite instruments such as the CERES EBAF data set [12, 19].

We use the equations from Soden et al. [29] to relate decadal change in CRE to equilibrium climate sensitivity (ECS) as defined in IPCC [12]. Let R_f denote the total anthropogenic radiative forcing of climate change by greenhouse gases, aerosols, and land change. T_s is global average surface temperature, and λ is climate sensitivity. Following Soden et al. [29]:

$$\Delta R_{\mathrm{f}}/\Delta T_{\mathrm{s}} = \lambda = \lambda_{\mathrm{p}} + \lambda_{\mathrm{L}} + \lambda_{\mathrm{w}} + \lambda_{\alpha} + \lambda_{\mathrm{csw}} + \lambda_{\mathrm{clw}}. \quad (1)$$

Note that $\Delta R_{\mathrm{f}}/\Delta T_{\mathrm{s}}$ and all λs are expressed in units of watts per square meter per Kelvin. The feedbacks are as follows:

λ_{p} Plank temperature feedback ~ -3.2
λ_{L} Temperature lapse rate feedback ~ -0.6

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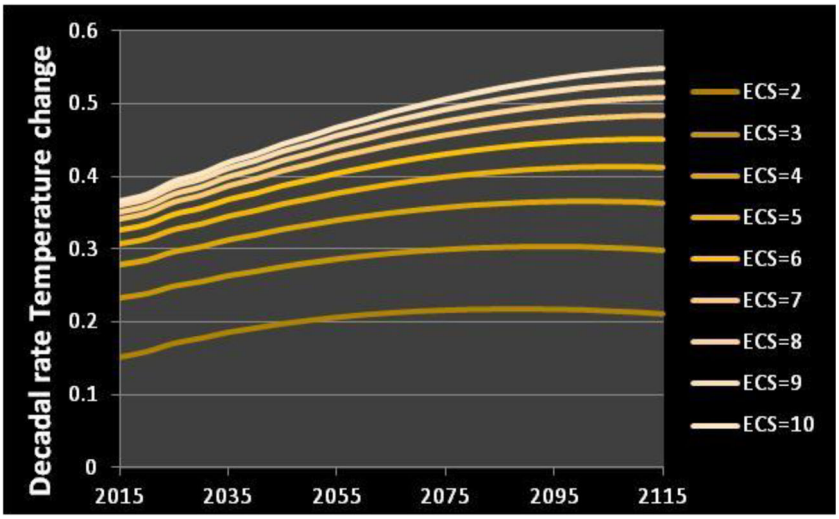


Fig. 1 Transient response of decadal change of global surface temperature (GST) (°C) for different ECS values, as function of year under the BAU emission path

λ_w Water vapor feedback ~ + 1.6

λ_α Snow and ice surface albedo feedback ~ + 0.3

λ_{csw} Shortwave cloud feedback = $\Delta CRE/\Delta T_s$. We vary λ_{csw} to develop the relationship between varying climate sensitivity and decadal change observations of ΔCRE.

λ_{clw} Longwave cloud feedback (not given separately in the IPCC report; using [28], Fig. 3 top, and averaging for all 12 of the climate models they used) ~ + 0.35

Positive magnitude is a positive feedback, and negative magnitude is a negative feedback.

We use estimates from the IPCC AR5 report, chapter 9, Figure 9.43, and Table 9.5, CMIP5 mean (red dot in the figures) for everything except the LW cloud feedback, which is not given in the IPCC report.

$$\lambda = \lambda_p + \lambda_L + \lambda_w + \lambda_\alpha + \lambda_{csw} + \lambda_{clw} \quad (2)$$

Solving for λ_{csw} with the values above,

$$\lambda_{csw} = \lambda - (-3.2) - (-0.6) - (+1.6) - (+0.3) - (+0.35) = \lambda + 1.55 \quad (3)$$

λ is simply related to the equilibrium climate sensitivity (ECS), as used in the Dynamic Integrated Climate–Economy (DICE) model, where ΔCO_2 denotes a doubling of atmospheric CO_2 concentration:

$$\lambda = \Delta R_f/\Delta T_s = (\Delta R_f \text{ for } \Delta CO_2)/(\Delta T_s \text{ for } \Delta CO_2) = -3.7/\text{ECS} \quad (4)$$

ECS in this definition is the amount of equilibrium global average surface temperature increase for an anthropogenic radiative forcing equivalent to a doubling of CO_2. The factor 3.7 converts a doubling of atmospheric CO_2 to watts per square meter of radiative forcing. See IPCC [12] for a discussion of the definition of radiative forcing. The idea here is that we set all of the feedbacks except SW cloud feedback equal to their average over the climate models. We then vary SW cloud feedback to obtain the range of climate sensitivity.

Combining (2), (3), (4), and $\lambda_{csw} = \Delta CRE/\Delta T_s$, the governing equation for the change in CRE is

$$100\Delta\text{CRE}(em, t, \text{ECS})/50 = 2\,\lambda_{csw}\,\Delta T\,(em, t, \text{ECS}) = 2\left[-3.7/\text{ECS} - \left(\lambda_p + \lambda_L + \lambda_W + \lambda_a + \lambda_{clw}\right)\right] \times \Delta T \quad (5)$$

In this equation, 50 is the global mean value of CRE_{sw} in units of watts per square meter and is used to convert the trend in watts per square meter into a trend in units of a fraction. A factor 100 converts fractions to percentages, resulting in the factor 2. We then have the decadal trend in shortwave cloud radiative effect in units of percent per decade. ΔT is determined by emissions scenario *em*, time *t*, and ECS. ΔT(*em*, *t*, ECS) is computed from DICE. Hence, the RHS is known and we can compute the theoretical value of ΔCRE(*em*, *t*, ECS) based on the DICE model, supplemented with Soden et al. [29]. ΔT represents the "true" global mean temperature change under these assumptions. Parenthetically, we note that Roe Baker adopted by IWGSCC use ECS = 1.2/(1 − f), $f \sim$ Normal(0.62, 0.19^2), whereas Soden et al. [29] used effectively $f \sim$ Normal(0.62, 0.1766^2). This difference is negligible.

If $\lambda_p \ldots \lambda_{clw}$ are uncertain, there is a case to be made for including this uncertainty. Is it plausible that, say, ECS = 10 should be attributed solely to λ_{csw}? Including these uncertainties would introduce complications, and in an initial study the gain in accuracy would not be compensated by the loss of clarity.

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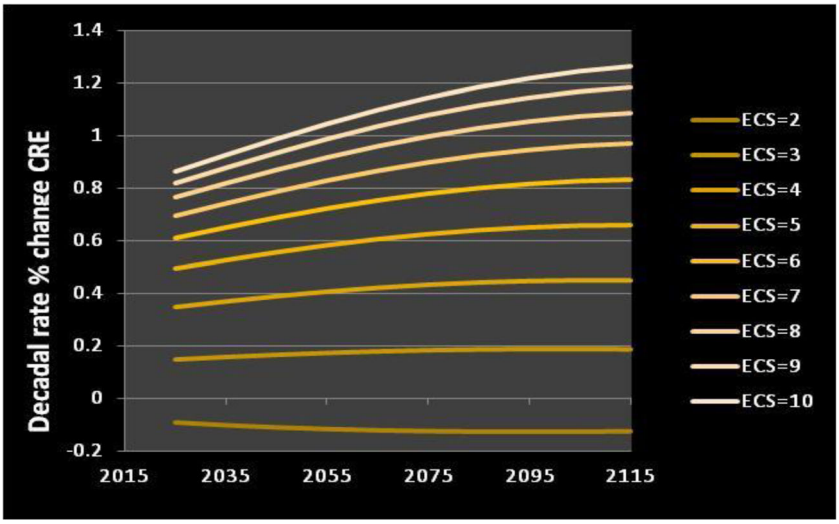


Fig. 2 Transient response of percentage change in cloud radiative effect (CRE) for different ECS values as function of year under the BAU emission path

Comparing Figs. 1 and 2, it appears that the GST rise paths are more closely packed for higher values of ECS than the decadal rate of percentage change in CRE. In 2025, the difference in CRE values for ECS = 3 and ECS = 2 is 0.24. The difference in CRE values for ECS = 10 and ECS = 9 is 0.046. For GST, these differences are 0.08 (ECS 3 − ECS 2) and 0.008 (ECS 10 − ECS 9). This could suggest to the unwary that it is easier to resolve the values of ECS, especially higher values, from measurements of CRE than from measurements of GST. However, such conclusions must take into account the additional noise of natural variability (not included in Figs. 1 and 2), measurement uncertainties for each, and also the probability of the different ECS values. These issues are addressed in the following section.

4 Observational Uncertainty

This section briefly reviews the statistics of observing time series. Suppose the physical variable of interest, X, is observed at evenly spaced time points. As depicted in Fig. 3, the estimate of the linear rate of change of X changes as new observations is added. We are interested in the uncertainty in the estimates of the rate of change. Let ΔX denote the linear rate of change in X per unit time. In Fig. 3, ΔX is estimated at 0.9 after four measurements; after the fifth measurement, the estimate of ΔX is 3.31.

The uncertainty in ΔX depends on the uncertainties of the individual observations, on the correlations between these measurements, and on the observation period. Following Wielicki et al. [31], the uncertainties of the individual observations from Earth Observing Systems are decomposed into independent contributions from natural variability in the climate system (*nat*), instrument calibration uncertainty (*cal*), and orbital sampling uncertainty (*orbit*). Natural variability is driven primarily by the internal nonlinear dynamics of ocean and atmosphere in the climate system. Calibration uncertainty (Leroy et al. 2008; [15, 32]) can be caused either by slow drifts of instrument calibration over years in orbit or by differences in absolute calibration between successive instruments separated by data gaps. Orbital sampling error can be caused either by limited space/time sampling or by systematic drifts in local time of day sampling for satellite instruments ([15]; IPCC, 2013). Although natural variability, calibration

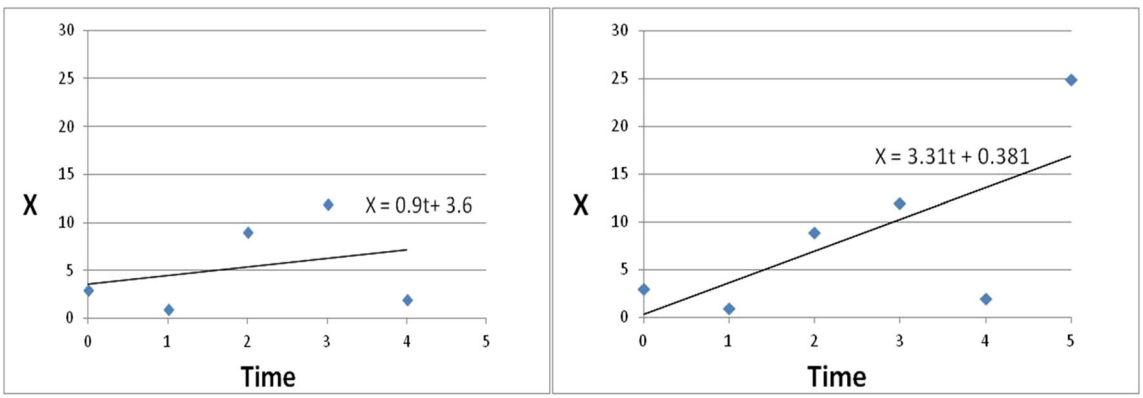


Fig. 3 Illustration of observing a trend in time series data

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Table 1 Uncertainty components for enhanced measurement of CRE and GST

	Enhanced CRE	Current CRE	Enhanced GST	Current GST
σ_{Nat}	0.6%	0.6%	0.085 K	0.085 K
τ_{Nat} (years)	0.8	0.8	2.3	2.3
σ_{cal}	0.15%	1.0%	0.03 K	0.18 K
τ_{cal} (years)	10	10	5	5
σ_{orbit}	0.21%	0.006%	0.018 K	0.018 K
τ_{orbit} (years)	1	1	1	1

uncertainty, and orbital sampling uncertainty are mutually independent, each has an autocorrelation time scale. This time scale has the effect of retarding the rate at which successive measurements drive down the variance of the trend. Leroy et al. (2008) derive the following expression for the variance in ΔX after observing the process for Δt time periods (a derivation is given in the Supplementary Online Material of [2]):

$$VAR(\Delta X) = 12(\Delta t)^{-3}\left(\sigma^2{}_{\mathrm{Nat}}\tau_{\mathrm{Nat}} + \sigma^2{}_{\mathrm{cal}}\tau_{\mathrm{cal}} + \sigma^2{}_{\mathrm{orbit}}\tau_{\mathrm{orbit}}\right) \quad (6)$$

where Δt is the length of observation period in years, $\sigma^2{}_{\mathrm{Nat}}$ is the one period variance of natural variability, and τ_{var} is the autocorrelation time scale of natural variability, similar terms apply for cal and orbit. If X is global surface temperature (C), then ΔX has dimensions (C/t) and $VAR(\Delta X)$ has dimensions (C^2/t^2), which is also the dimension of the terms $(\sigma^2{}_i\tau_i)/\Delta t^3$.

The values used in Eq. 6 are taken from Wielicki et al. [31] and are shown in Table 1. To convert to decadal rates of change, the square root of Eq. 6 is multiplied by 10 thereby converting uncertainty per year to uncertainty per decade. As in the earlier papers, the calibration time scale is taken as the nominal time between instrument launches which has been longer for research mission CRE observations (~ 10 years) than for weather satellite GST observations (~ 5 years).

Recall from Eq. 1 that $\lambda_{\mathrm{csw}} = \Delta\mathrm{CRE}/\Delta T_s$. In this case, observations of SW cloud feedback incorporate both uncertainty in CRE and temperature trends. In principle, both can be included in the analysis. From Fig. 4 below, the standard deviation of CRE dominates that of T by roughly a factor 6 for both the current and enhanced systems. Accounting for the change of units from percentage to watts per square meter reduces this to a factor 3. A standard error analysis shows that to first order, the uncertainty of λ_{csw} is dominated by the uncertainty in CRE. Adding temperature trend uncertainty to the cloud feedback uncertainty would not modify the conclusions of the paper, while substantially increasing the complexity of the analysis and the explanation of results.

An examination of Table 1 suggests the need to understand the relative accuracy of CRE or GST trend observations. Figure 4 shows the standard deviations of trends from each measurement as a function of year and shows that the enhanced observing system improvement for CRE (right) is larger than that for GST (left).

Both enhanced systems greatly reduce uncertainty relative to the current systems. The key question, however, is the degree to which the various systems are able to resolve the transient climate response signals shown in Figs. 1 and 2. The difference between adjacent response signals changes with time and with the value of ECS. To appreciate the subtleties involved, Fig. 5 compares the ratios of trend standard deviations to the differences in response signals for ECS = 4 − ECS = 3 (left) and for ECS = 10 − ECS = 9 (right).

Figure 5 shows that the enhanced GST system has slightly higher resolving power for ECS = 3 than the enhanced CRE system, but the roles are reversed for ECS = 9. In all cases, the enhanced CRE system offers greater improvement over its current system than does the enhanced GST system.

Supposing that we can implement only one of these two enhanced systems, the choice comes down to the social benefit we expect from each system.[2] Following Cooke et al. [1–3], we propose to quantify the benefit of each system by considering each as a real option and comparing their real option value (ROV) with respect to their respective current systems. This answers the question "is it better to upgrade the GST measurement system, or to upgrade the CRE measurement system?" Problems of combining signals for decision making, while more realistic, are considerably more complex and are best handled from a fully Bayesian framework—a subject of future work (for early results, see [10]).

Since information can have value only if it is used, computing the ROV involves positing a decision context in which each system would be used to trigger societal decisions. Of course, such social decisions may be triggered on combinations of signals.

[2] There are some differences in our assumptions: we assumed 10-year instrument lifetime (tau cal) for the CRE obs and only 5 years for the GST. This means that calibration errors decrease in time more slowly for CRE than for GST. We did this to be consistent with the previous papers and because the weather system tends to update its instruments more often on orbit than research climate data. If the world were to implement a designed climate observing system, then this would likely change to 5 years for CRS as well.

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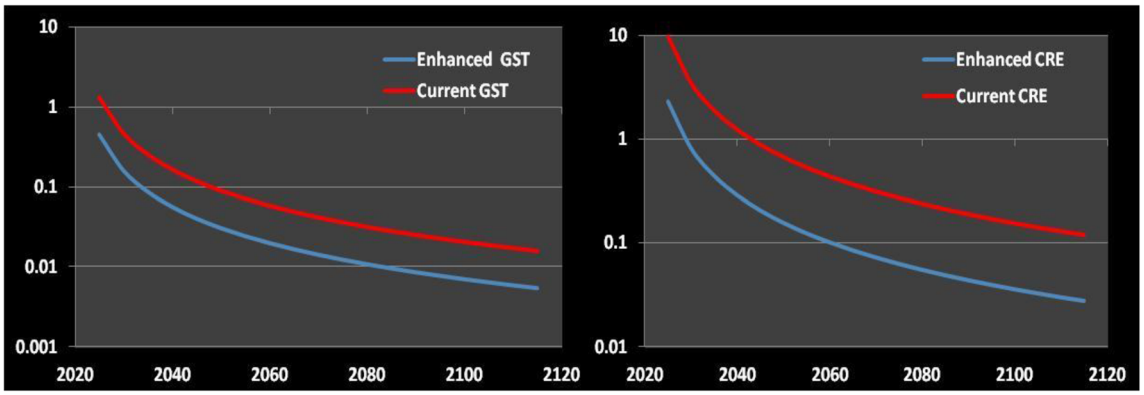


Fig. 4 Standard deviations of observed trends for enhanced measurements of GST (left figure) and CRE (right figure) as function of year from launch for enhanced platforms (blue) and current platforms (red) [31]

5 Decision Context

The decision context invoked in Cooke et al. [1, 2] is applied here. In this context, society follows a BAU emission path until a trigger variable (in this case either CRE or GST) is observed to exceed a trigger value with requisite confidence. At that point, society switches to one of three reduced emission paths. These are termed the "DICE Optimal" (this is the optimal emission path if ECS is known to take the value 3C), the "lim 2.5" (this path leads to stabilizing global surface temperature at 2.5 °C above the pre-industrial level, assuming ECS = 3), and the "Stern" path which pursues much more aggressive emissions reduction. These emission paths are shown in Table 2.

As in Cooke et al. [1, 2], we posit a launch date, a trigger value, and a confidence level. When the triggering variable is observed to exceed the trigger value with the given confidence level, society switches to one of three reduced CO_2 scenarios. For the enhanced and the current EOS, the expected net benefits are calculated using the DICE integrated assessment model [24, 26], for discount rates 2.5%, 3%, and 5% using the truncated Roe Baker distribution for ECS [14]. Net benefits are the PV of damages averted by switching emissions scenarios minus the PV of abatement costs. The ROV of the enhanced EOS is the surfeit expected net benefits of triggering the switch on the more accurate enhanced EOS instead of the less accurate current EOS.

A base case involves a 95% requisite confidence level and trigger values of 0.2C/decade (GST) and − 0.1%/decade (CRE). The trigger values are chosen to induce the switch when we become 95% certain that ECS is greater than 2C. The value of 2C is chosen as representative of climate sensitivity sufficiently high to drive policy changes. Higher climate sensitivity requires larger emissions reductions in order to limit the level of future climate change. The linkage between the trigger values and ECS = 2 is only approximate, owing to the curvilinear behavior in Figs. 1 and 2.

The base case in this study uses 2020 as the nominal start of the climate change observations used to constrain uncertainty in climate sensitivity. There are two reasons for this. First, a CLARREO Pathfinder satellite mission capable of improving the calibration of CERES in orbit has a planned launch date of 2020 on the International Space Station. Second, large uncertainty remains in anthropogenic aerosol indirect radiative forcing [12].

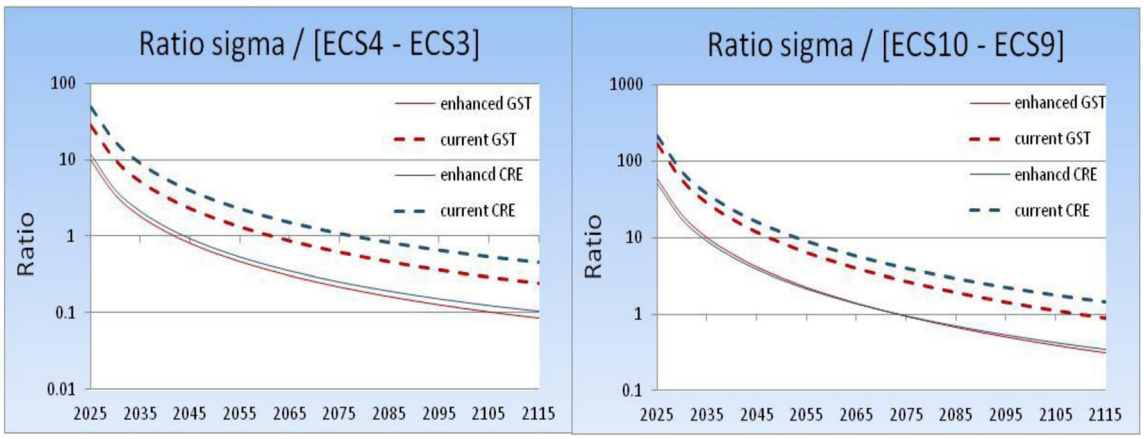


Fig. 5 Ratios of standard deviations to the differences in response lines for ECS = 4 − ECS = 3(right) and for ECS = 10 − ECS = 9 (right)

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Table 2 Emission pathways

	Total carbon emissions (GTC per year)											
	2005	2015	2025	2035	2045	2055	2065	2075	2085	2095	2105	2115
BAU	9.058	10.463	12.395	14.566	16.741	18.716	20.388	21.699	22.593	23.158	23.361	22.640
DICE Opt	9.058	8.956	9.994	10.838	11.227	11.027	10.222	8.887	7.149	5.154	3.044	0.932
Lim2.5C	9.058	8.897	9.868	10.576	10.716	10.106	8.702	6.601	4.079	1.684	0.541	0.401
Stern	9.058	5.200	4.974	4.653	4.211	3.630	2.878	1.951	0.805	0.148	0.118	0.0945

Indirect aerosol forcing modifies cloud properties and therefore can be confused with cloud feedback changes in X. Both direct and aerosol indirect forcing uncertainties also dominate the uncertainty in using past temperature trends to narrow uncertainty in ECS [12]. This remains a very active research area, and we assume here that the enhanced EOS will have developed improved aerosol indirect effect observations by 2020. Finally, while aerosol radiative forcing is expected to reduce in magnitude in the future as China and India improve air pollution controls and coal use is reduced, CO_2 continues to build in the atmosphere with a very long lifetime, thereby becoming the dominant anthropogenic forcing in the next few decades. These considerations all suggest that 2020 is the earliest date to realistically consider starting the VOI comparison of an enhanced EOS versus the current EOS.

The choice of the reduced emission path is made at the moment the trigger is "pulled," based on the triggering observation. Hence, each EOS may be regarded as a real option for choosing a new emission path (or indeed, retaining BAU) at the uncertain moment in the future when the trigger is pulled. The net benefits of triggering under an EOS are computed as expected net present value of the averted damages (relative to damages under the BAU) minus the costs of abatement. For the enhanced measurement of GST, net benefits are computed in Cooke et al. [1]. Enhanced measurements of CRE are computed in Cooke et al. [3].

The trigger time is computed using a simple Gaussian error model. If OBS(ΔX) is the observed value of the linear rate of change of X when the true value is ΔX, then we assume OBS(ΔX) + ξ = ΔX where ξ is normal with mean zero and variance σ^2. To be 95% certain that $\Delta X > \alpha$, we must have

$$\begin{aligned} 0.95 < P(\Delta X > \alpha) &= P(\mathrm{OBS}(\Delta X) + \xi > \alpha) \\ &= P(\xi/\sigma > (\alpha - \mathrm{OBS}(\Delta X))/\sigma) \\ &\Leftrightarrow (\alpha - \mathrm{OBS}(\Delta X))/\sigma < -1.65 \hat{U} \mathrm{OBS}(\Delta X) \\ &> \alpha + 1.65\sigma \end{aligned} \tag{7}$$

Here, 1.65 is the 95th percentile of the standard normal variable ξ/σ. The standard deviation is the root of Eq. (6). The parameter α is called the trigger value of ΔX, and 1.65 is termed the confidence level or simply the confidence. Figure 6 shows the 90% central confidence bands around values of ECS = 2, 3, …10 for the enhanced CRE system. Of course we do not observe ECS; rather, we observe the values on the vertical axis.

Figure 7 illustrates triggering by showing only the 5% lower confidence bands for the enhanced and current

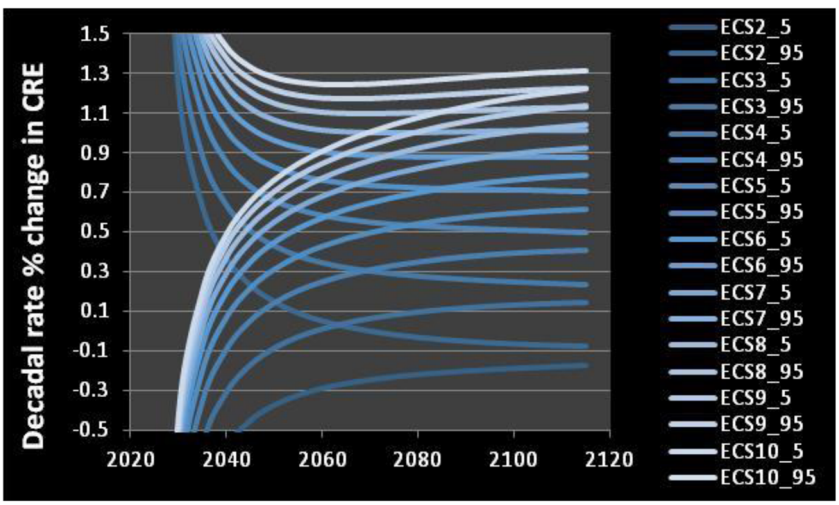


Fig. 6 For the enhanced CRE system, 5 and 95% confidence bands for values of ECS = 2, 3, … 10. ECS2_5 denotes the 5th percentile of % change in CRE as function of time, if ECS = 2; ECS2_95 denotes the 95th percentile

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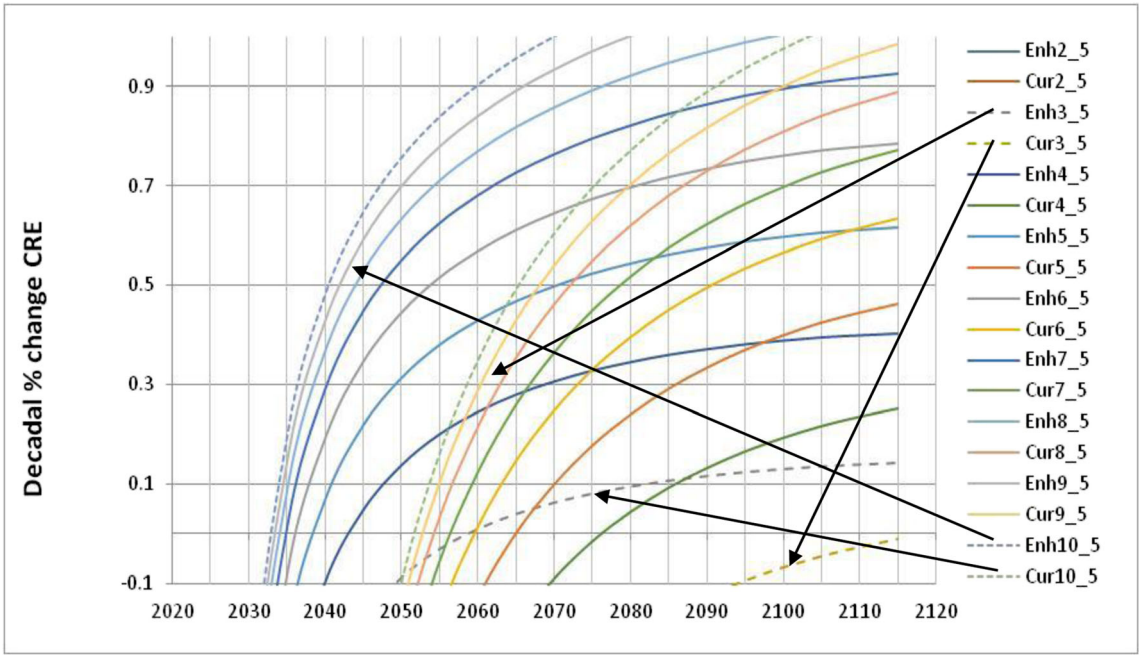


Fig. 7 Trigger times for enhanced and current EOSs for CRE, triggering at 95% certainty of exceeding − 0.1. The dashed lines show the ECS = 3 and 10 cases considered in Fig. 5

CRE EOSs. The trigger value of − 0.1 intersects the horizontal axis. The point at which a 5% confidence band for ECS = 3C hits the horizontal axis denotes the year in which the trigger would be pulled if the true unknown value of ECS were actually 3. The enhanced system would yield 95% confidence that ECS ≥ 3C in the year 2048. With the current system, triggering would happen in 2090 if ECS = 3C. Figure 8 shows the same information for GST. For GST, if ECS = 3C, triggering occurs at 2040 and 2060 for the enhanced and current EOSs, respectively. For visual convenience, the lines corresponding to the latest trigger times in the graph are dashed and the lines corresponding to earliest triggering (highest ECS) are dotted. The enhanced trigger times in Fig. 7

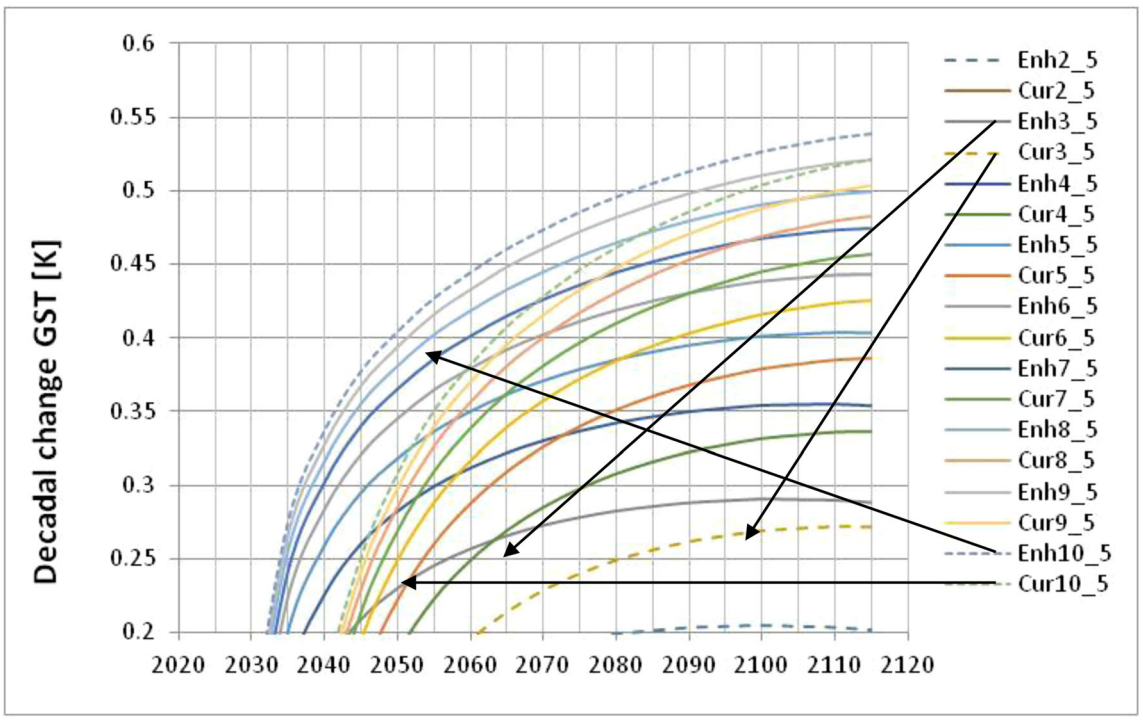


Fig. 8 Trigger times for enhanced and current EOSs for GST, triggering at 95% certainty of exceeding 0.2

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are a bit later than those of Fig. 8. However, the trigger times for the current CRE are much later than those for the current GST. This means that the improvement effected by the enhanced CRE system should be greater than that achieved by the enhanced GST system.

Note that the enhanced and current systems do not trigger at all for ECS = 2C.

The calculations compute the decadal rates of change at 5-year time steps. Triggering could occur in year 2045 only for 5% confidence bands intersecting the horizontal before 2045 and after 2040 (otherwise, they would trigger in 2040). The current system cannot trigger before 2050; the enhanced system cannot trigger before 2030.

Notice that the effects on triggering times of raising the trigger value, say to 0.25 K for GST, are easily read off from the points at which the 5% confidence lines cross the horizontal corresponding to 0.25.

6 Real Option Value of Enhanced CRE and Enhanced GST

As stated above, information can have value only if it is used. Any calculation of value of information or real option value must therefore posit a decision context in which the information would be used. The decision context is not a prediction of how social choices will be made, but a tool for quantifying the value of information. It may be compared to a thought experiment in physics in which an idealized situation is used to reveal underlying structure.

To introduce the calculations, we first disable the real option feature of choosing the reduced emission path after triggering. Rather, upon triggering, the reduced emission path is "hard wired" to each of the three paths in Table 2. Figure 9 shows the different expected net benefits as a function of ECS, for enhanced and current CRE EOSs. For each value of ECS, we consider the triggering times for the enhanced and current

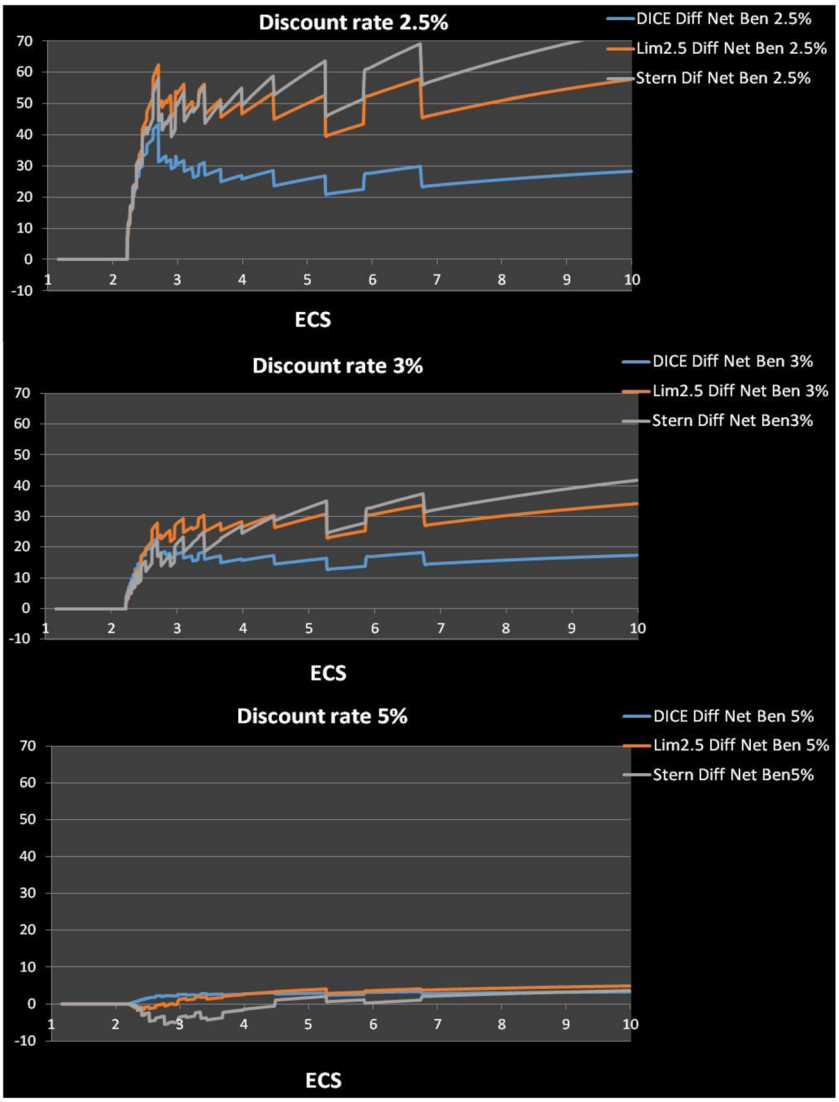


Fig. 9 Expected surfeit net benefits (trillion USD 2008) from three reduced emissions scenarios for different ECS values and different discount rates, trigger value − 0.1, confidence 95%, for CRE

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systems (these times are different) and upon triggering switch to each of the reduced emission paths.

The calculation is repeated for discount rates 2.5%, 3%, and 5%. The jerky behavior is caused by the discretization to 5-year time steps. For example, for ECS = 4C, the enhanced CRE system triggers at 2040 and the current system at 2060. As ECS increases, the damages increase, but not the abatement costs, and the differences with the current CRE also increases. This continues until we reach the higher ECS value at which either the enhanced system triggers at 2035 or the current system triggers at 2055, at which point the surfeit expected net benefits change discontinuously. Thus, the graph shows the amount by which a switch to each emission path is improved by triggering on the enhanced versus the current

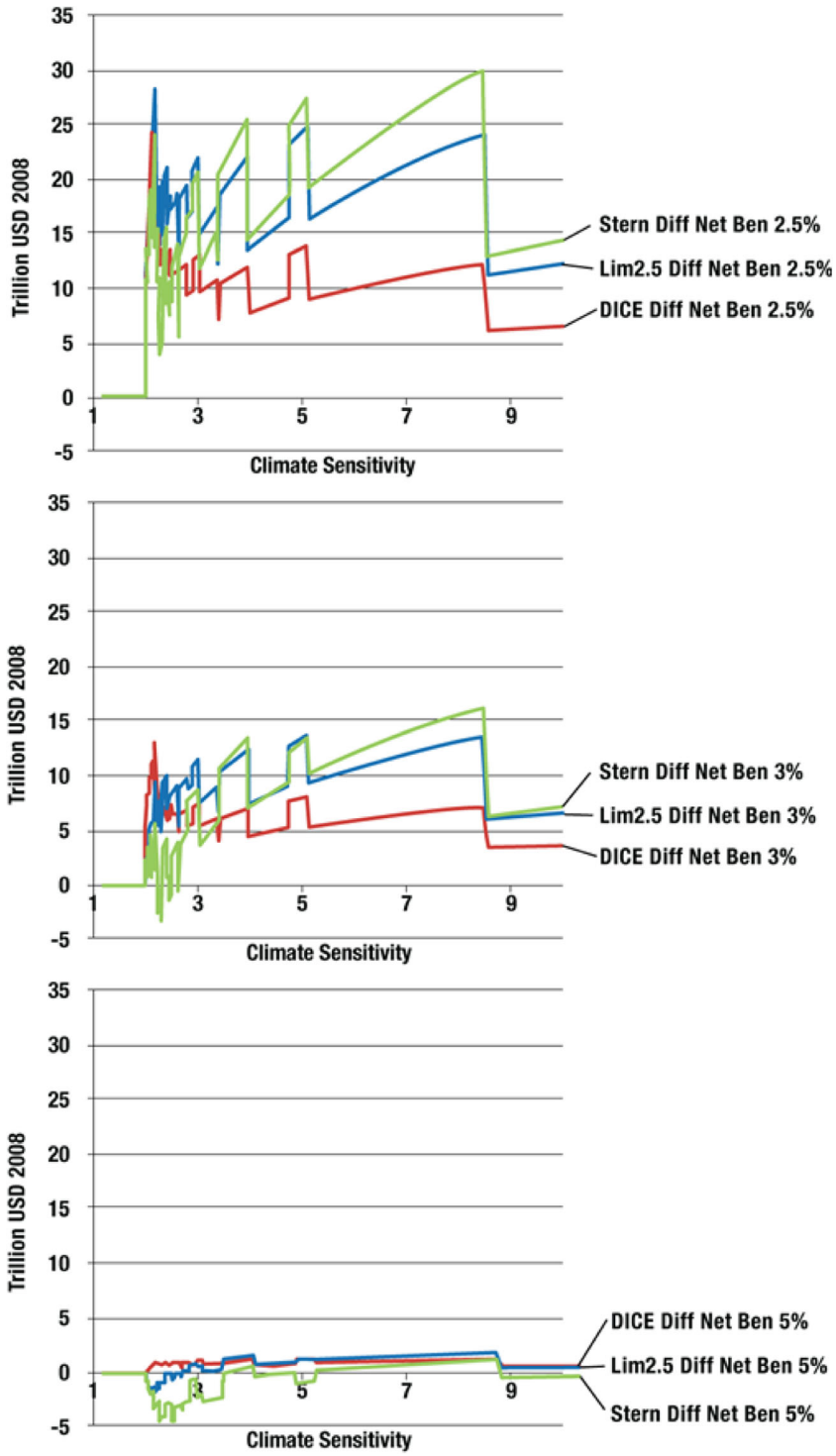


Fig. 10 Surfeit net benefits (trillion USD 2008) for enhanced EOS (triggering on change in temperature rise) versus current EOS as functions of ECS. The horizontal axis is (unknown) climate sensitivity; the vertical axis is surfeit benefits as a function of climate sensitivity. Jumps are caused by the 5-year discretization (from [1]).

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Table 3 Real option value of enhanced CRE and enhanced GST measurements

Enhanced systems real option value: surfeit expected net benefits relative to current systems (trillion USD 2008)

	GST			CRE		
Trigger value	0.2C			− 0.1		
Confidence	95%			95%		
Launch year	2020			2020		
Discount rate	2.5%	3%	5%	2.5%	3%	5%
	16.7	9	1.07	38.88	20.08	2.00

system, for all values of ECS. It must be emphasized that the surfeit expected net benefits does not take into account the damages suffered under BAU before any switching occurs.

The DICE optimal path (which is optimal for ECS = 3 and 5% discounting) has marginally better surfeit expected net benefits than Lim 2.5 only with discount rate 5% and ECS ≤ 4.6. With 2.5% discount rate, the Stern path is optimal at all values of ECS, and for 3%, it is optimal for ECS above 5. The surfeit expected net benefits of the Stern path are negative for 5% discounting for ECS < 4.5C. It is better to take a bad decision later rather than earlier.

Figure 10 (from [1]) shows analogous information to Fig. 9 for triggering on decadal GST rise. Note that the scale on the vertical axes reflects the higher values of surfeit net benefits for CRE relative to GST. The negative values for Stern for the GST systems for 5% discount rate are echoed in Fig. 10.

Reinstating the real option feature, we allow that upon triggering, the emission path with the highest net expected benefits is chosen. The expectation is computed with respect to the distribution of ECS values, conditional on triggering at the given time.

Given the 5-year step size, triggering in a given year defines an interval of admissible values of ECS, as depicted in Figs. 7 and 8. At each 5-year time step, the expected net benefits are averaged over the admissible ECS values and the reduced emission path with optimal expected net benefits is defined. When the enhanced system triggers in a given year, the optimal expected net benefits in that trigger year are decremented by the optimal expected net benefits for the later years in which the current system would trigger. The result is the surfeit expected net benefits of the enhanced system relative to the current system.

The major conclusion is that the surfeit expected net benefits of the enhanced versus the current EOS are larger when triggering on decadal change CRE (Table 3) than when triggering on decadal change in temperature (Table 3).

7 Conclusions

This paper demonstrates how real option theory in conjunction with the social cost of carbon can be used to compute the real option value of alternative ways of learning about equilibrium clinate sensitivity, namely measuring global surface temperature and cloud radiative effect. Owing to the greater resolving power of the cloud radiative effect measurements, and the greater improvement above existing systems, the real option value of the cloud radiative effect measurement system is roughly double that of the global surface temperature measurement system. Hence, if budgets enable the launch of only one of these systems, the results of this paper indicate a strong preference for cloud radiative effect. It is perhaps not surprising that the net benefits of CRE are found to be much higher than for GST. Several authors (e.g., [6]) have shown that the largest uncertainties in estimating ECS arise from uncertainties in CRE. Thus, a new EOS that substantially reduces the uncertainty in CRE would be expected to have more value than learning GST alone. Simply put, the new EOS greatly reduces the overall uncertainty in Eq. 1 by substantially reducing its largest uncertainty term. Furthermore, changes in CRE (i.e., *the cause*) result in changes in GST (i.e., *the effect*), and hence, learning *why* an effect occurs should have more value than observing the effect itself. In this case, changes in CRE are directly related to the uncertain cloud feedback that is driving uncertainty in climate sensitivity. Use of temperature trends is much less direct. It is critically dependent on knowing the radiative forcing of greenhouse gases and aerosols in order to constrain climate sensitivity. Total human driven radiative forcing uncertainty remains a factor of 3 because of large uncertainties in aerosol forcing [12]. The current initial analysis does not capture these much more complex distinctions. Future studies should endeavor to address these broader issues as part of the value of information context.

It is natural to ask what is the real option value of launching both systems? The methodology deployed here is certainly feasible but its application would be much more complex. Instead of observing a linear trend with noise, we are observing a two-dimensional planar trend with noise. Instead of confidence intervals, we have confidence elipses. The variance decomposition in Eq. 6 would have to be revisited. The decision context would become less intuitive; instead of a target variable and target value, we should have to speak of ordered pairs of target variables and of values. Undoubtedly, our real problem is much more complex still, and working out the details for two dimensions would be a worthy, albeit taxing, endeavor.

Bayesian Nets (BNs) provide graphical tools for visualizing complex interactions of disparate measuring systems, and visualizing how much newer enhanced systems add to what we will already learn from exisiting systems, taking account of the uncertainty in the underlying physical processes. Simple

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intuitions based on elementary statistics of independent measurements on which most scientists have been raised can and will lead us far astray. These must be addressed and sorted out before tackling more complex higher dimensional decision problems. Some early results of the Bayesian Net approach can be found in Hanea et al. [10].

While future improvements in the methodology can be made, the current analysis clearly demonstrates the very large potential economic value of developing a designed climate observing system. In this context, a factor of 3 increase in the global investment in climate observations, analysis, and modeling is predicted to lead to a return on investment of roughly 50 to 1 (Cooke et al. 2013). With this large return, factors of 2 or even 5 uncertainty in the value of information do not alter the fundamental conclusion.

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